OTT Platform Management System

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Project Explanation (Overview)

- This project is a **console-based OTT platform system** developed using Java.
- It allows users to manage movie data and perform various operations.
- The system uses **arrays to store movie details** like ID, title, genre, and rating.
- It provides an interactive **menu-driven interface** for user input.

Project Explanation (Core Concepts Used)

- **Arrays:** To store movie details (ID, Title, Genre, Rating)
- **Stack:** Used to maintain **watch history (LIFO)**
- **Queue:** Used for **watch later list (FIFO)**
- **Loops & Switch Case:** For menu-driven execution
- **Scanner Class:** To take user input

Project Explanation (Features)

- Add new movies with details
- Display all available movies
- Watch a movie (stored in history using Stack)
- View watch history
- Add movies to watch later (Queue)
- View watch later list
- Search movie by ID
- Sort movies based on rating

Stack (Watch History)

- **Purpose:**
- Helps find a specific movie quickly using ID
- Improves user experience by locating required data
- **Where Used in Code:**
- Method: searchMovie()
- Uses **Linear Search**
- Loop:
- for(int i=0;i<count;i++)
- Condition:
- if(movieId[i] == id)

Queue (Watch Later)

- **Purpose:**
- Queue follows **FIFO (First In First Out)** principle
- Stores movies to watch later in order
- Ensures movies are watched in the same order added
- **Where Used in Code:**
- Queue<String> watchLater = new LinkedList<>();
- watchLater.add(title); → adds movie to queue
- viewWatchLater() → displays queue elements

Searching

- **Purpose:**
- Helps find a specific movie quickly using ID
- Improves user experience by locating required data
- **Where Used in Code:**
- Method: searchMovie()
- Uses **Linear Search**
- Loop:
- for(int i=0;i<count;i++)
- Condition:
- if(movieId[i] == id)

Sorting

- **Purpose:**
- Arranges movies based on ratings
- Helps users identify top-rated movies easily
- **Where Used in Code:**
- Method: sortByRating()
- Uses **Bubble Sort Algorithm**
- Condition:
- if(rating[j] < rating[j+1])
- Swapping:
  - Ratings
  - Movie ID
  - Title
  - Genre

OTTPlatform.java X *OTTPlatformDemo.java

```java
package p1;
import java.util.*;
public class OTTPlatform {

        static Scanner sc = new Scanner(System.in);

        // Arrays to store movie data
        static int[] movieId = new int[100];
        static String[] movieTitle = new String[100];
        static String[] genre = new String[100];
        static double[] rating = new double[100];
        static int count = 0;

        // Stack for watch history
        static Stack<String> watchHistory = new Stack<>();

        // Queue for watch later
        static Queue<String> watchLater = new LinkedList<>();


        // Add movie
        static void addMovie() {
            System.out.print("Enter Movie ID: ");
            movieId[count] = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Movie Title: ");
            movieTitle[count] = sc.nextLine();

            System.out.print("Enter Genre: ");
            genre[count] = sc.nextLine();

            System.out.print("Enter Rating: ");
            rating[count] = sc.nextDouble();

            count++;

            System.out.println("Movie added successfully.\n");
        }


        // Display movies
```

OTTPlatform.java *OTTPlatformDemo.java X

```java
package p1;

import java.util.Scanner;

public class OTTPlatformDemo {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your choice: ");

    int choice;

    do {

        System.out.println("===== OTT PLATFORM MENU =====");
        System.out.println("1. Add Movie");
        System.out.println("2. Display Movies");
        System.out.println("3. Watch Movie");
        System.out.println("4. View Watch History");
        System.out.println("5. Add to Watch Later");
        System.out.println("6. View Watch Later");
        System.out.println("7. Search Movie by ID");
        System.out.println("8. Sort Movies by Rating");
        System.out.println("9. Exit");
        choice = sc.nextInt();

        switch(choice) {

            case 1:
                OTTPlatform.addMovie();
                break;

            case 2:
                OTTPlatform.displayMovies();
                break;

            case 3:
                OTTPlatform.watchMovie();
                break;
```

Console

OTTPlatformDemo [Java Application] C:\Users\chint\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre

```
Enter your choice: ===== OTT PLATFORM MENU =====
1. Add Movie
2. Display Movies
3. Watch Movie
4. View Watch History
5. Add to Watch Later
6. View Watch Later
7. Search Movie by ID
8. Sort Movies by Rating
9. Exit
3
Enter movie title to watch: annabell

Movie added to watch history.

===== OTT PLATFORM MENU =====
1. Add Movie
2. Display Movies
3. Watch Movie
4. View Watch History
5. Add to Watch Later
6. View Watch Later
7. Search Movie by ID
8. Sort Movies by Rating
9. Exit
1
Enter Movie ID: 28474
Enter Movie Title: conguring
Enter Genre: horror
Enter Rating: 4.7
Movie added successfully.

===== OTT PLATFORM MENU =====
1. Add Movie
2. Display Movies
3. Watch Movie
4. View Watch History
5. Add to Watch Later
6. View Watch Later
7. Search Movie by ID
8. Sort Movies by Rating
```

Thank You

- Thank You for Your Time
- Any Questions?